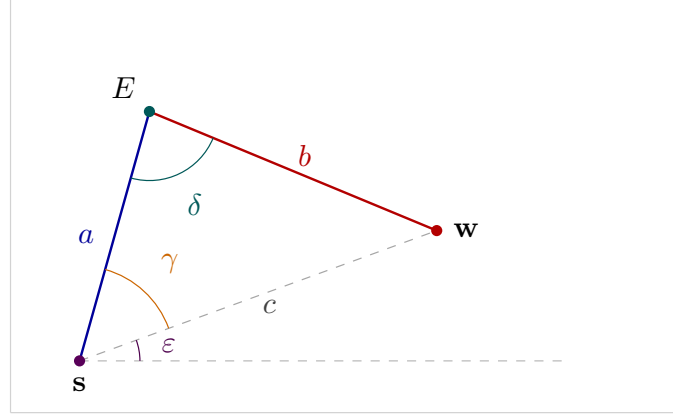


IKCompiler.calc_shoulder_and_elbow_angles()



Physical links sE (length a) and Ew (length b), solid. Auxiliary straight-line chord sw (length c) and world-horizontal reference, dashed. γ, δ solved by the law of cosines on triangle sEw ; ε is the chord's own elevation, so $\gamma + \varepsilon$ is link a 's elevation in the world frame.

1 Setup

$$\begin{aligned} s &= \text{shoulder_servo_position}, & w &= \text{wrist_position} & \in \mathbb{R}^3 \\ a &= \text{shoulder_arm_length}, & b &= \text{elbow_arm_length} \end{aligned}$$

E is the (unknown, never explicitly computed) elbow position. It is fixed only implicitly, as the point at distance a from s and distance b from w , in the vertical plane containing s , w , and the z -axis.

2 Elevation Angle

$$\mathbf{d} = \mathbf{w} - \mathbf{s}, \quad r = \sqrt{d_x^2 + d_y^2} \quad (1)$$

$$\varepsilon = \text{atan}\left(\frac{d_z}{r}\right) \quad (2)$$

the angle the chord sw makes above the world-horizontal plane (atan suffices in place of atan2 since $r \geq 0$ always).

$$c = \|\mathbf{d}\| = \sqrt{r^2 + d_z^2} \quad (3)$$

3 Law of Cosines

Triangle sEw has sides $a = sE$, $b = Ew$, $c = sw$. The law of cosines gives its interior angles at s and E :

$$\gamma = \angle(s) = \arccos\left(\frac{a^2 + c^2 - b^2}{2ac}\right) \quad \text{angle between } sw \text{ and } sE \quad (4)$$

$$\delta = \angle(E) = \arccos\left(\frac{a^2 + b^2 - c^2}{2ab}\right) \quad \text{angle between } Es \text{ and } Ew \quad (5)$$

$\delta = 180^\circ$ is the fully-extended (straight) configuration s, E, w collinear – the reference the servo-space elbow command is built against ($\text{elbow_logical_zero} + (180 - \text{elbow_angle})$).

4 World-Frame Shoulder Angle

γ is measured from the chord **sw**, not from horizontal. Since ε is exactly the chord's own elevation, adding it rotates γ into the world frame – the elevation of link a itself:

$$\boxed{\text{shoulder_angle} = \gamma + \varepsilon} \quad \text{elbow_angle} = \delta \quad (6)$$

5 Notation $\leftrightarrow$ Code

Symbol	Variable
s	<code>shoulder_servo_position</code>
w	<code>wrist_position</code>
a, b	<code>shoulder_arm_length</code> , <code>elbow_arm_length</code>
d	<code>shoulder_2_wrist_vec</code>
r	<code>xy_projection_dist</code>
ε	<code>shoulder_2_wrist_vec_angle</code>
c	<code>shoulder_2_wrist_length</code>
γ	(<code>shoulder_angle</code> , before the += correction)
δ	<code>elbow_angle</code>
$\gamma + \varepsilon$	<code>shoulder_angle</code> (returned)